# Project Submission–Database Administration

# STUDENT EXAMINATION DATABASE

## STUDENT DETAILS

| | |
|---|---|
| Student name | S. Pavithra |
| Register no | 2422j0629 |
| Department | BCA |
| ID | 5E6D2E95B3565DFBECE4B728E46E5342@VTPT2026.in |
| College | Tiruppur Kumaran college for women |
| Lastdateof submission | 25/09/2026 |
| Github link | |

## 1. Pro blem Statement & Objectives – 5 Marks

### Problem statement

Createa database for maintaining examination details, subjects, students, marks and results. The database should organize student and subject information, store examination records and marks, calculate grades, and generate a student

## Objectives

• M•aintain student details in a structured database.A vehicle service center needs a database to efficiently
• Store and manage subject details.
• Maintain examination records for students and subjects information about customers, their vehicles, and the services performed on
• Store marks and calculate grades automatically.
those vehicles. Maintaining these details manually makes it difficult to track
• Provide easy insertion, updating, deletion and retrieval of records.
service history, retrieve customer information, and calculate service-related
• Generate student result reports using SQL queries and joins.
• Maintain data accuracy using primary keys, foreign keys, NOT NULL and UNIQUE constraints.costs. The Vehicle Service Management Datbase stores this information in related tables and allows the service center to perform data insertion,
updating, deletion, searching, filteri

## 2. DatabaseDesign –10Marks

### 2.1 Database Structure

TheStudent Examination Database is designed to store student,subject,examination,marksand result information in related tables. Primary keys uniquely identify records and foreign keys establish relationships between the tables

### Relationships

- One student can have many examination/marksrecords.
- One subject can appear in many examination/marks records.
- One examination can contain marks for many students and subjects.
- Each marks record can be used to generate a corresponding grade/result.

### Table used

| Table name | Purpose |
|---|---|
| Student | Stores student ID, name, department and year |
| Subject | Stores subject ID, subject name, maximum marks and credits. |
| Exam ination | Stores examination ID, examination name and examination date. |
| Marks | Stores student marks for each subject and examination |
| Result | Stores grade and result status calculated from marks |

### 2.2 Relationship Summary

| From | Relationship | To |
|---|---|---|
| Student | 1:M | Marks |
| Subject | 1:M | Marks |
| Examination | 1:M | Marks |
| Marks | 1:1 | Result |

## 2.2 Entity Relationship Diagram

The ER diagramshowsthemaintables,attributes,primary keys, foreign keys and relationships used in the Student Examination Database.

| STUDENT | MARKS | SUBJECT | EXAMINATION | RESULT |
|---|---|---|---|---|
| StudentID (PK) | MarkID (PK) | SubjectID (PK) | ExamID (PK) | ResultID (PK) |
| StudentName | StudentID (FK) | SubjectName | ExamName | MarkID (FK) |
| Department | SubjectID (FK) | MaxMarks | ExamDate | Grade |
| Year | ExamID (FK) | Credits | | ResultStatus |

### 2.3 Technologies Used

| Technology / Tool | Purpose |
|---|---|
| MySQL | Database management system used to store and manage student, subject, examination, marks and result record |
| MySQL Workbench | Used to create tables, execute SQL queries, manage the database and view query results. |
| SQL | Used for database creation, data manipulation, filtering, grouping, joins and result calculation. |

## 3. Practical Implementation –1.5Marks

### 3.1 Database and Table Creation

#### Datab aseCreation

```
CREATE
DATABASEstudent_exam_db;

USE
student_exam_db;
```

#### Student Table

```
Student (
Stude
                ntID
  INT PRIMARY
SKtEuYd,entName VARCHAR(100)
NOT NULL,
Department VARCHAR(50)
NOT NULL,
YearINT NOT
NULL );
```

#### Subject Table

```
CREATE TABLE Subject ( SubjectID
INT PRIMARY KEY, SubjectName
VARCHAR(100) NOT NULL
, MaxMarks INT NOT NULL,
Credits INT );
```

## 3.2 Constraints and Relationships

Thedatabaseusesprimarykeystouniquelyidentifyrecords,fore ignkeystoconnectrelatedtables, NOT NULL to ensure required values,and UNIQUEtoprevent duplicate values where necessary.

| Table | Primary Key | Foreign Key | Other Constraints |
|---|---|---|---|
| Student | Student id | - | NOT NULL |
| Subject | Student id | - | NOT NULL |
| Examination | Exam id | - | NOT NULL |
| Marks | Mark id | StudentID, SubjectID, ExamID | NOT NULL |
| Result | Result id | Mark id | NOT NULL |

## Marks Table

```
CREATETABLEMarks (
MarkIDINT PRIMARY KEY,
  StudentID INT NOT NULL,
ExamID INT NOT NULL,
Marks INT NOT NULL,
FOREIGN  KEY  (StudentID)  REFERENCEStudent(StudentID),
FOREIGN KEY (Subje ctID)  SREFERENCESubject(SubjectID),
FOREIGN KEY (ExamID)REFSERENCESExamination(ExamID)
);
```

## Result Table

```
ResultID INT PRIMARY     KEY,
MarkID INT NOT NULL,
Grade VA)RC;HAR(5),
ResultStatus VARCHAR(20),

FOREIGNKEY REFERENCE     Marks(MarkID)
(MarkID) );       S
```

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# 3.3 Data Insertion

## Stude ntData

```
INSERTINTOstudent VALUES
(1,'Arun Kumar','BCA',3),
((ı32,',ʀ’pɑvrıi sʏhɑɑdnkeɑvr,i’ʙ,’cbɑc,ɜɑj’,,3),
(4,’meena devi’,’bca’,3)
```

## Subj ect Data

```
INSERTINTOSu bject VALUES
(101,'Database Management Systems',100,4),
(102,'PythonProgramming',100,4),
(103,'Web Technology',100,4),
(104,'Software Engineering',100,4);
```

## ExaminationData

```
IN SERTINTOExamination VALUES
1001 , ‘semester examination ‘,2026-08-20’);
```

## Marks data

```
INSERTINTOMarks
VALUES (1,1,101,1001,86),
(2,1,102,1001,91),
(3,1,103,1001,88),
(4,1,104,1001,87),
(5,2,101,1001,78),
(6,2,102,1001,82),
(7,2,103,1001,76),
 (8,2,104,1001,82);
```

## 3.4 SQL operations

### INSERT

```
INSERT INTO student ALUES (5,’kartik raj’, ‘BCA’,3);
```

### SELECT

```
SELECT * FROM Student;
```

### UPDATE

```
UPDATE Student SET Department='BCA' WHERE StudentID=5;
```

### DELETE

```
DELETE FROM Marks WHERE MarkID=12;
```

### WHERE / ORDERBY

```
OSERLEDCTE *R F BROYM Marks WHERE Marks > 80;
SELECT * FROM Marks ORDER BY Marks DESC;
```

### LIKE / BETWEEN

```
SELECT * FROM Student WHERE StudentName LIKE 'A%';
SEL ECT * FROM Marks WHERE Marks BETWEEN 60 AND 90;
```

Theseoperationsareusedtoinsert,retrieve,modify,remove,filterand sort examination data.

# 3.5GradeCalculation and Student Result Report

## Grade Calculation

```
SELECTStudentID, SubjectID,Marks,
CASE
WHEN Marks >= 90 THEN 'A+'
WHEN Marks >= 80 THEN 'A'
WHEN Marks >= 70 THEN 'B'
WHEN Marks >= 60 THEN 'C'
ELSE 'F' END AS Grade
FROM Marks;
```

## GROUP BY / Average

```
SELECTStudentID,COUNT(*)AS Subjects, AVG(Marks) AS AverageMarks FROM Marks GROUP BY StudentID;
```

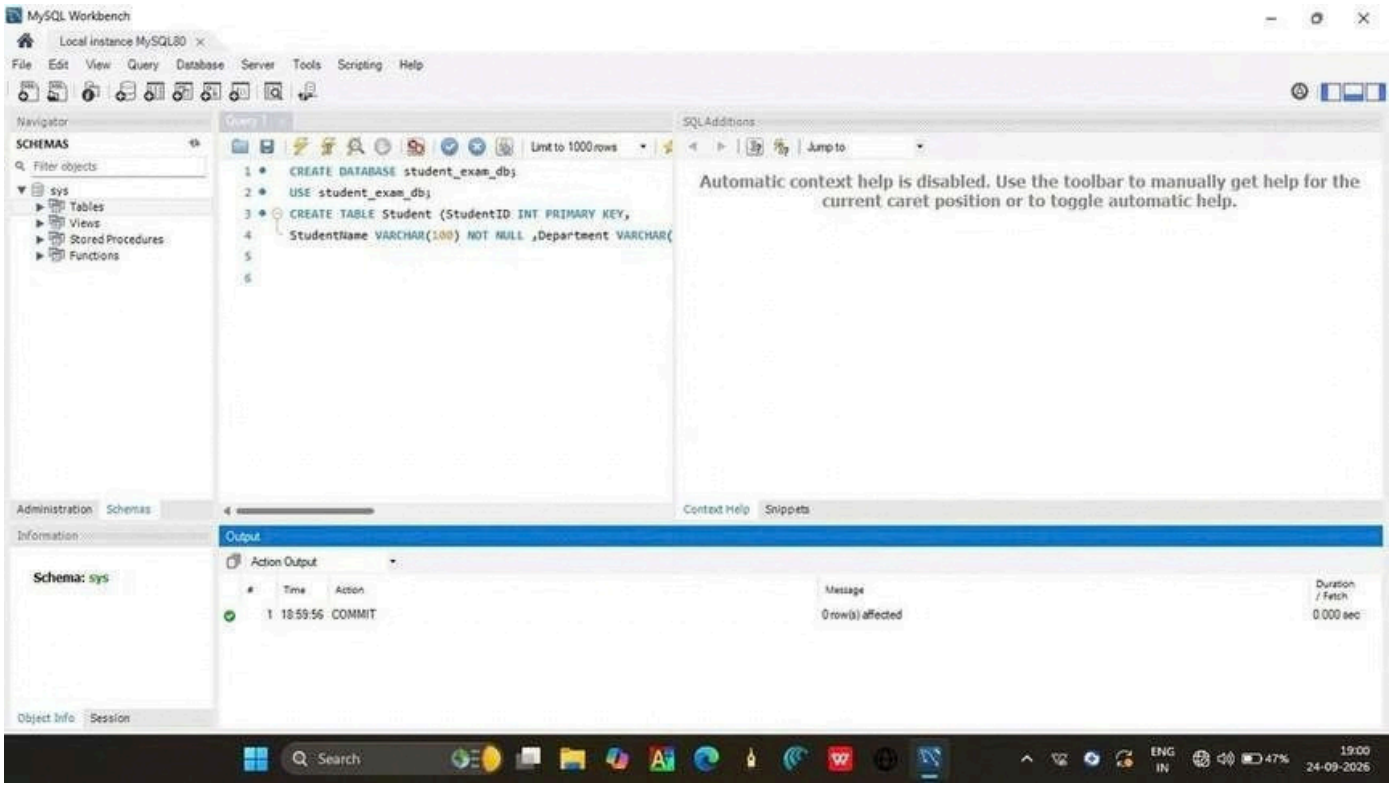
## Student Result Report

```
SELECT s.StudentName, sub.SubjectName, m.Marks,r.Grade,r.ResultStatus
FROM Student s
FROM Customer c INNER JOIN Marks m ON s.StudentID = m.StudentID INNER JOIN Subject
sub ON m.SubjectID = sub.SubjectID
INNER JOIN Vehicle v ON c.customer_id = v.customer_id
INNER JOIN Result r ON m.MarkID = r.MarkID;
```

The final report combines student, subject, marks and grade information so that the result can be viewed in one report

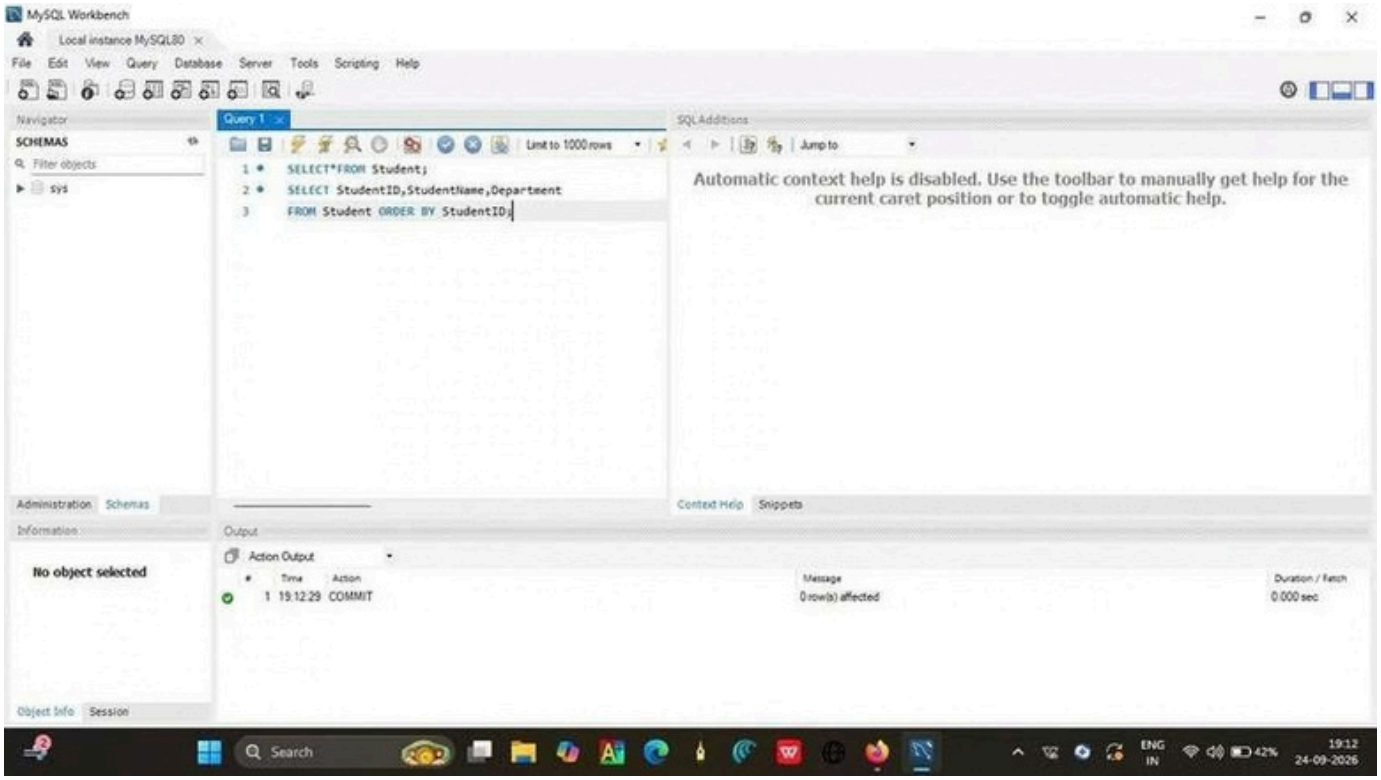
# 4. Output Screenshots–10Marks

Screenshot1–DatabaseandTables



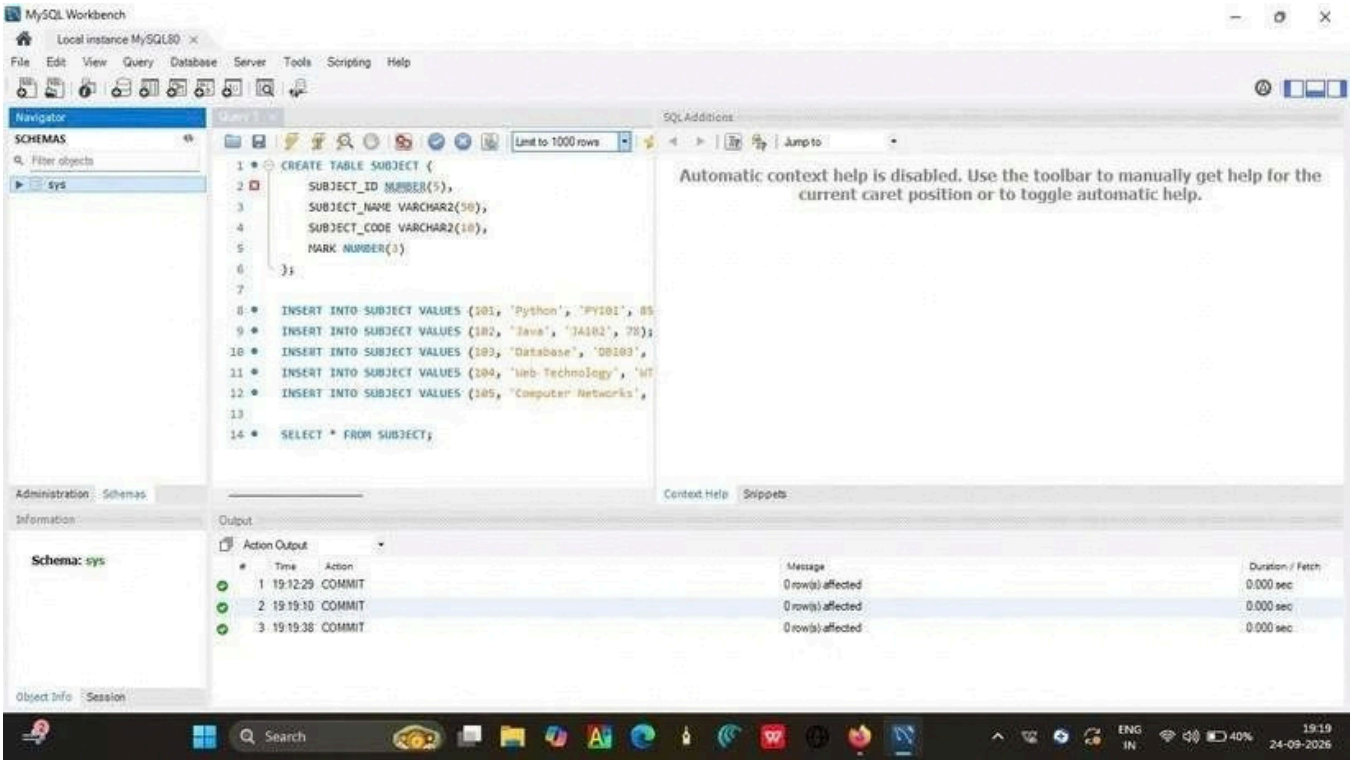
## 4. Output Screenshots – 10 Marks

### Screenshot 2 – Student Table with Records



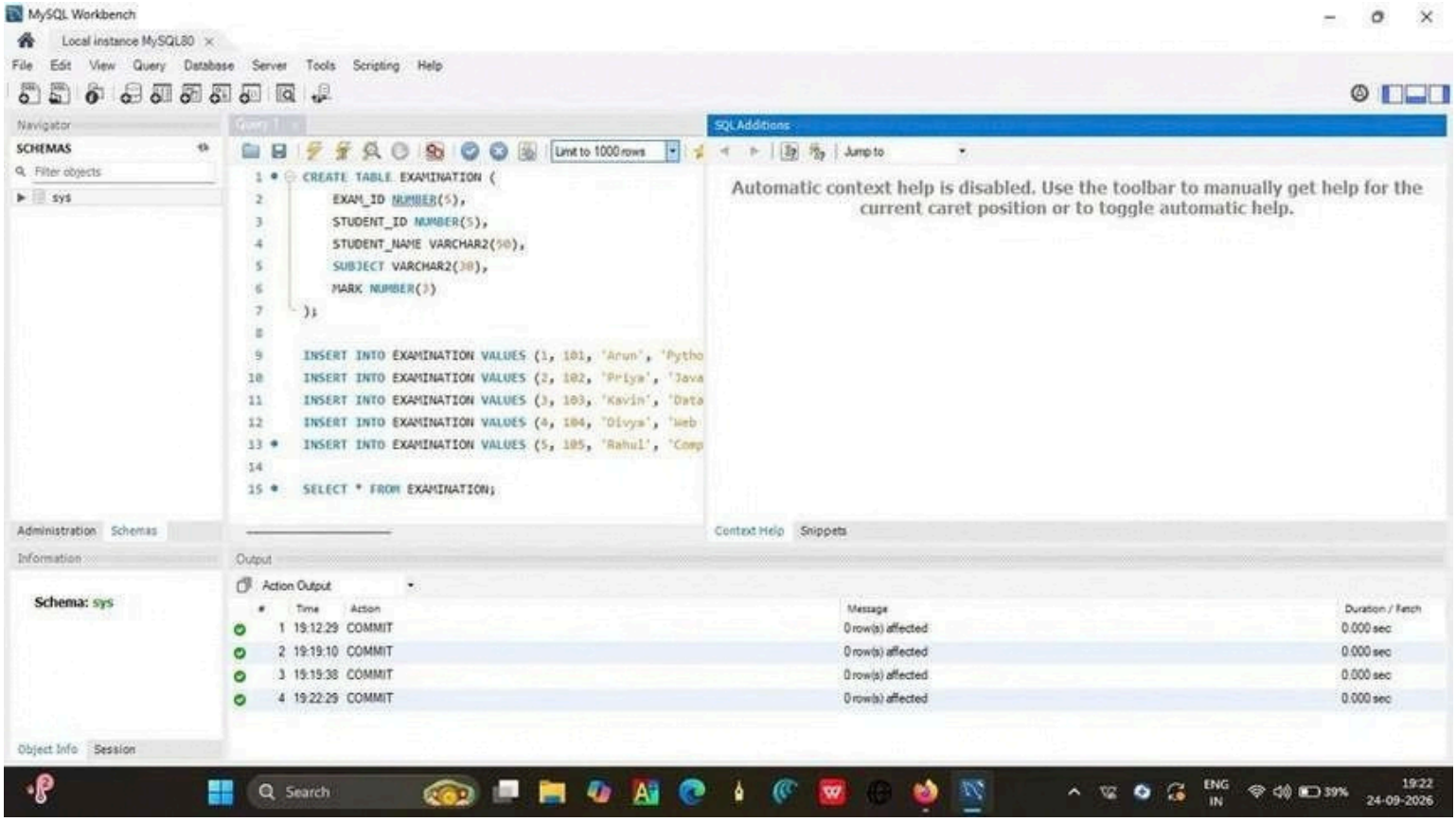
## 4.Output Screenshots-10 marks

### Screenshot 3 – Subject Table with Records



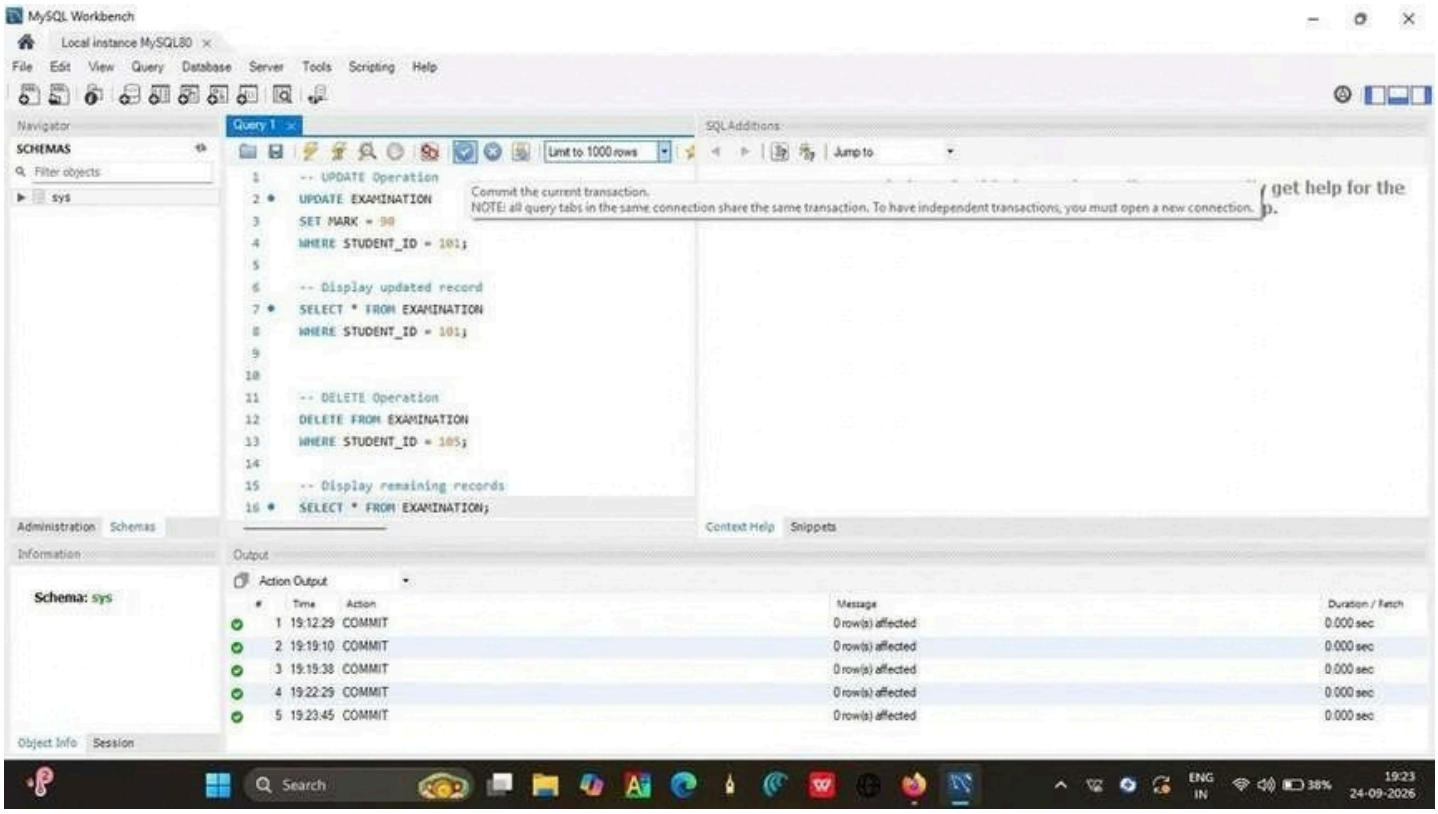
## 4.Output Screenshots - 10 Marks

### Screenshot 4 – Examination and Marks Records



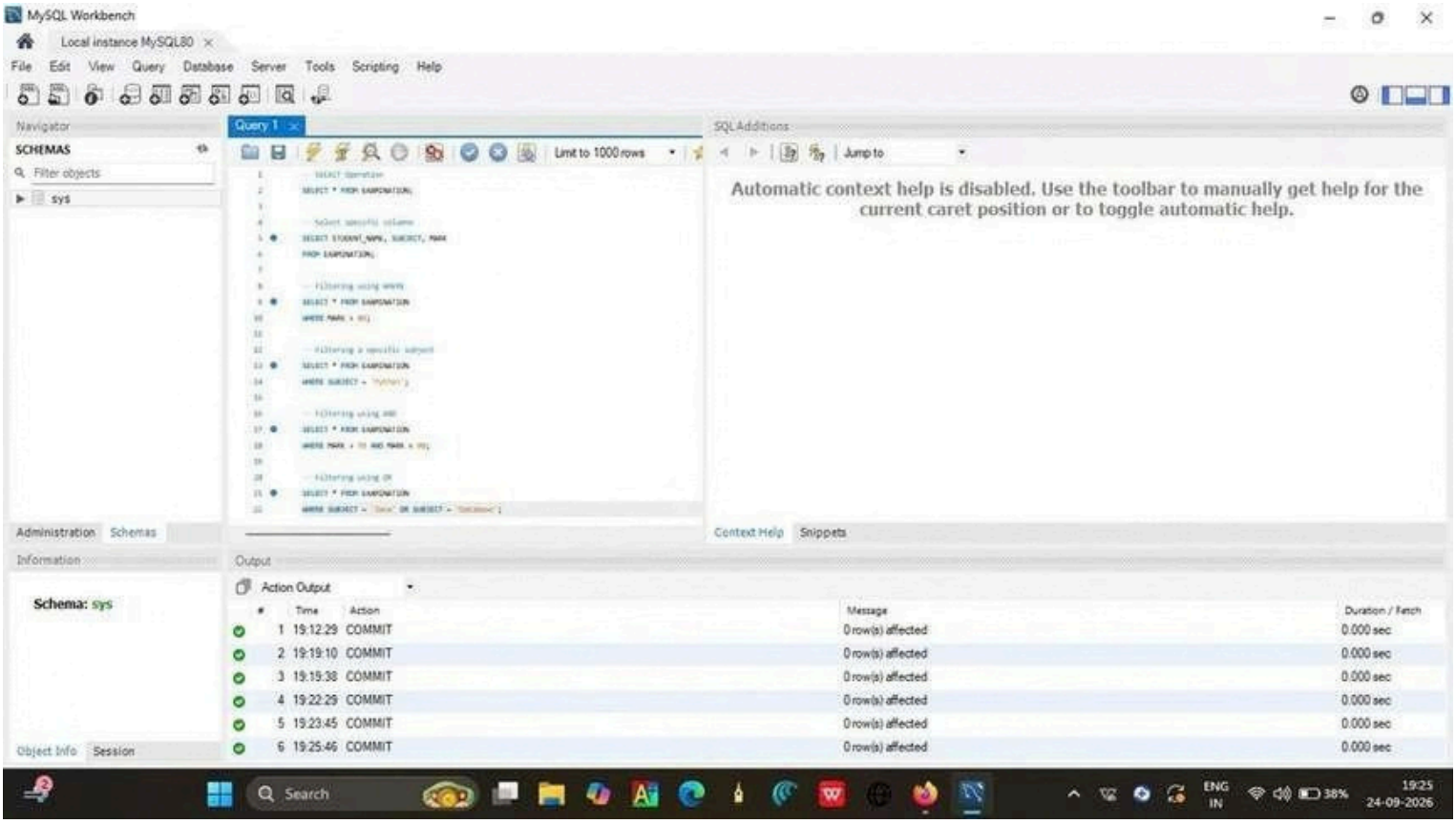
## 4.Output screenshots - 10 Marks

### Screenshot 5 – UPDATE/DELETE Operations



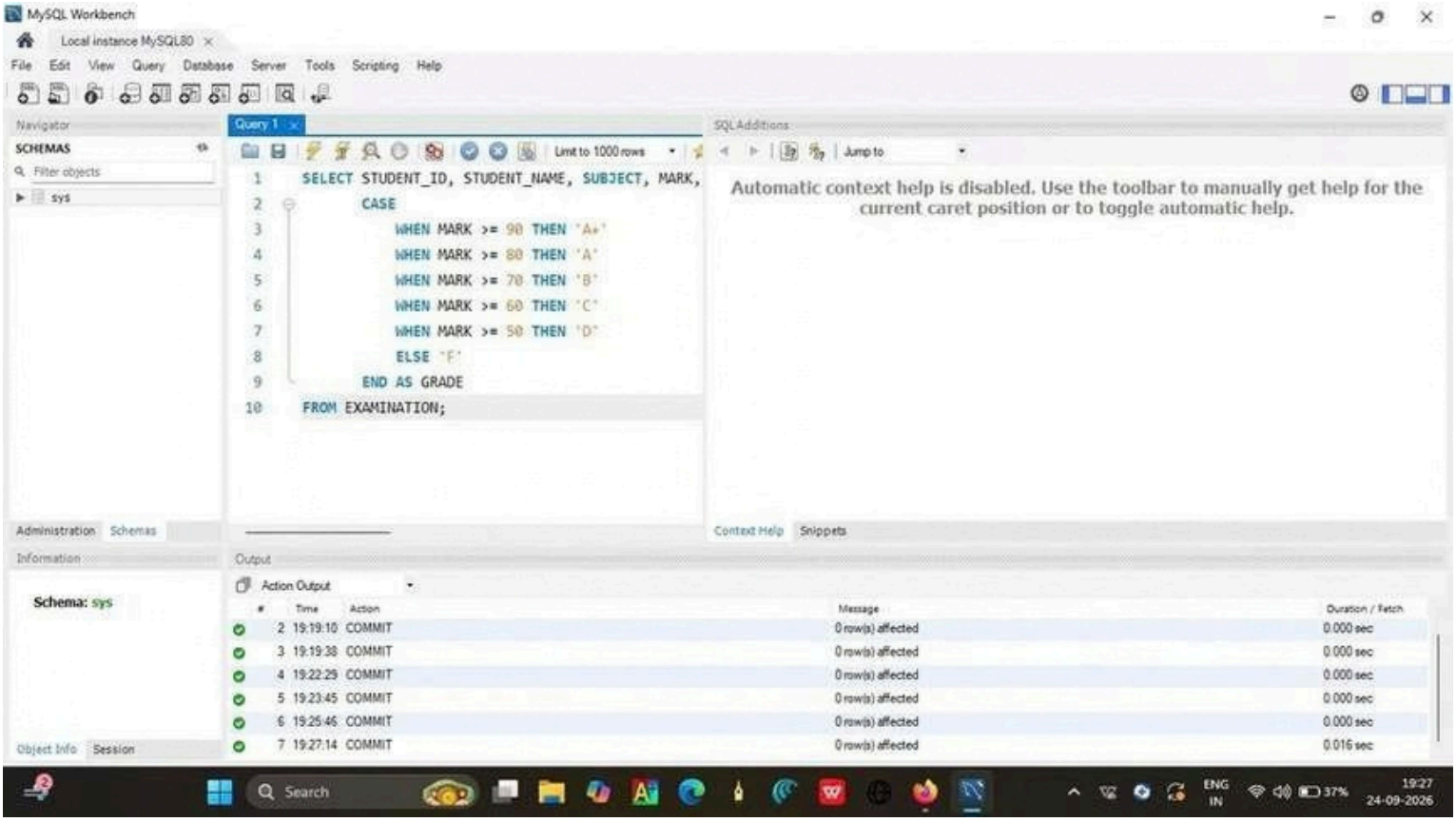
## 4. Output Screenshots – 10 Marks

### Screenshot 6–SELECTandFilteringOperations



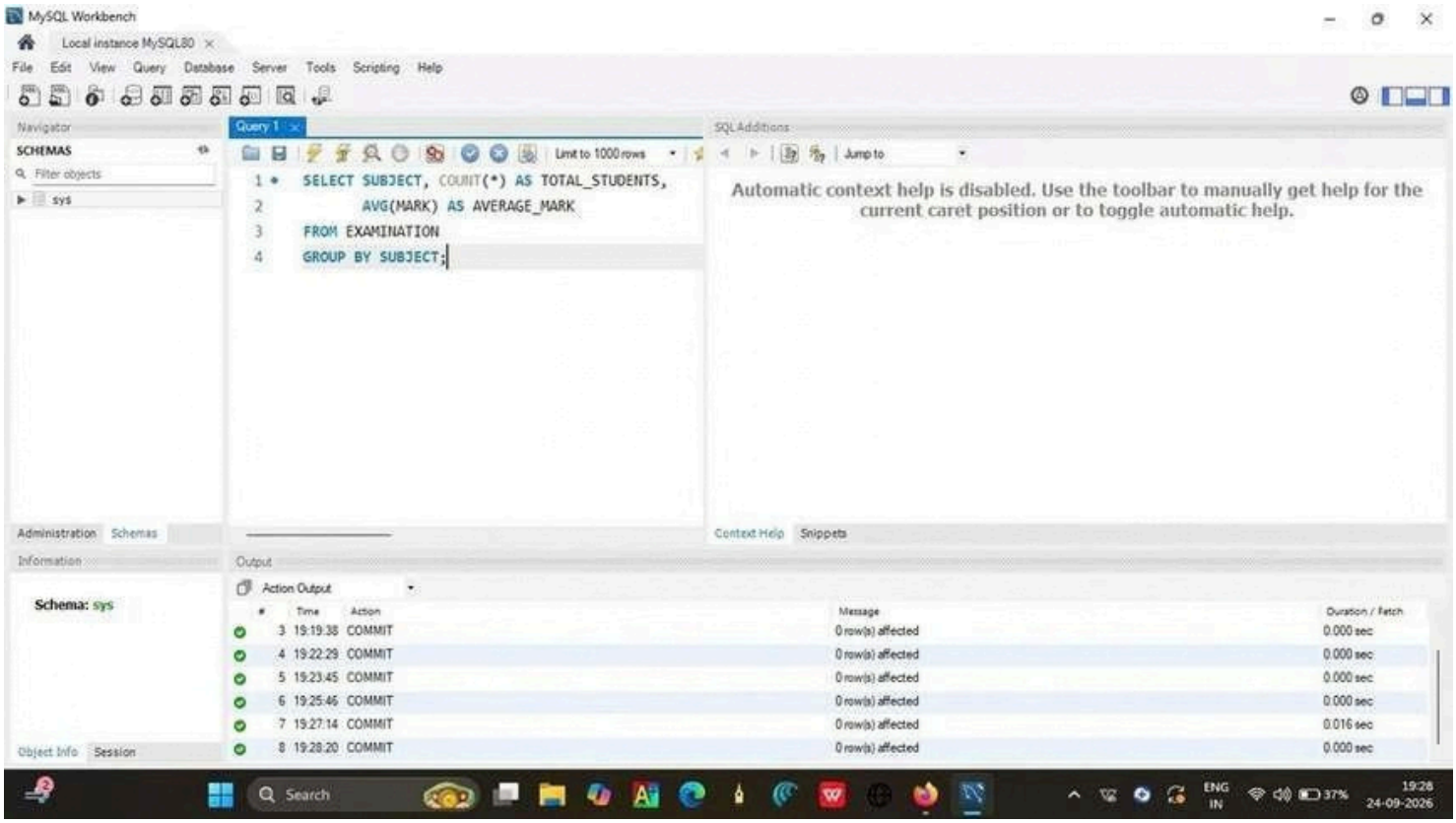
## 4. Output Screenshots – 10 Marks

### Screenshot 7 – Grade Calculation



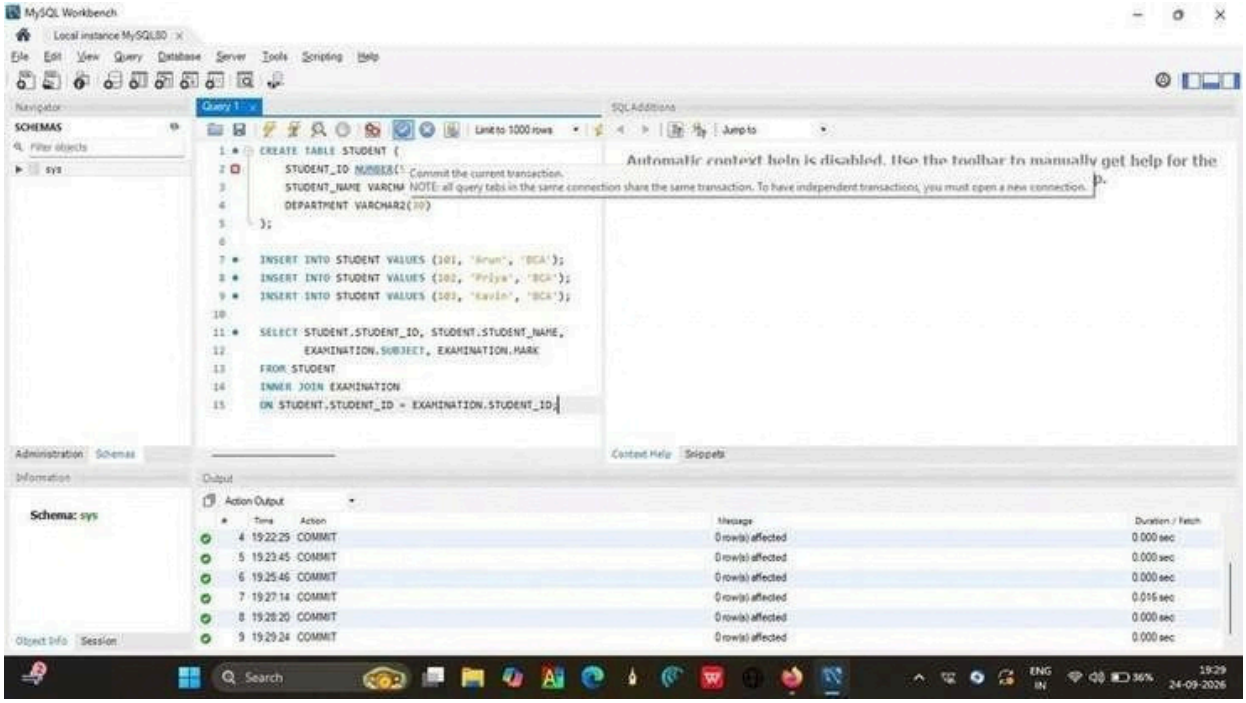
## 4. Output Screenshots – 10 Marks

### Screenshot8 –GROUP BY Operation

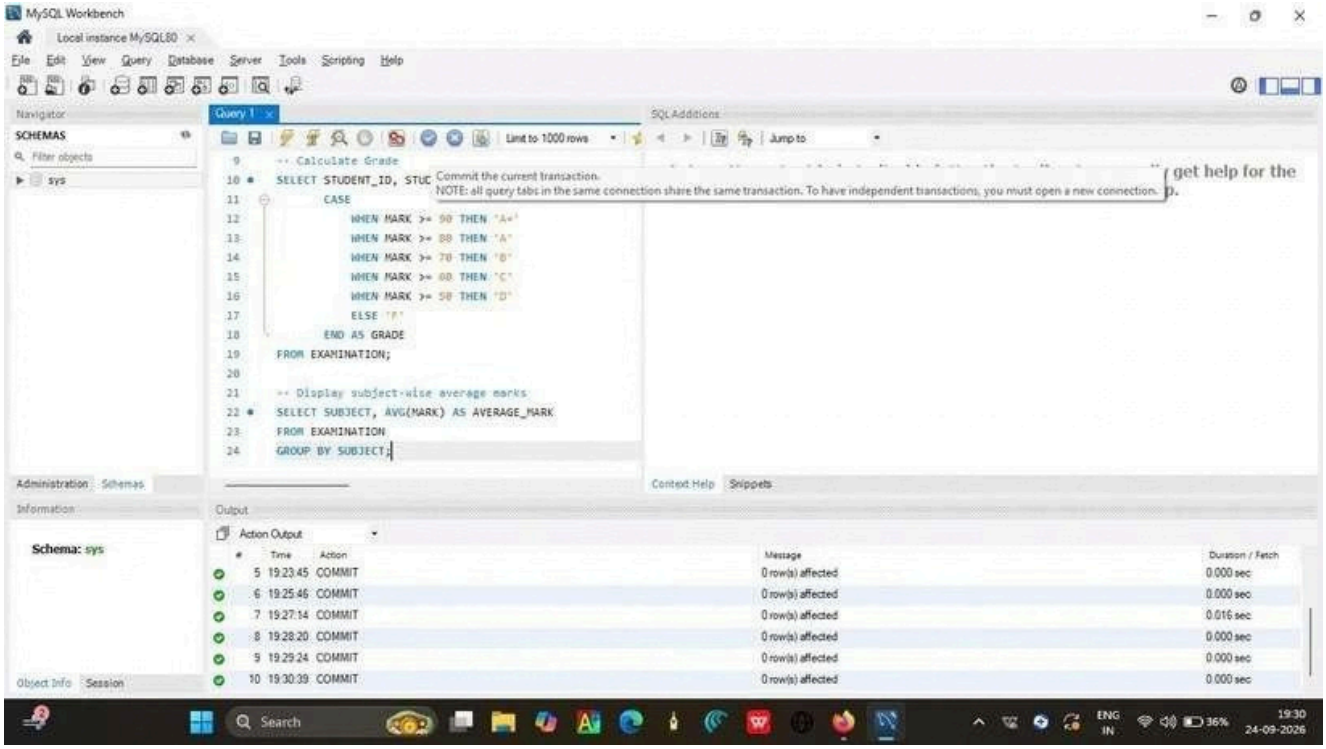


## 4.Output Screenshots -10 marks

### Screenshot9–INNERJOINOperation




### Screenshot10–FinalDatabase/QueryResults



## 5. Technical Explanation – 5 Marks

### Data Retrieval

WHEREfiltersrecordsbasedonacondition,ORDERBYsortsrecords,LIKE searches for partial values, BETWEENretrievesvalueswithinarange,andGROUPBYgroupsrecordsaccording to a selected column.

### SQL Operations

INSERT adds records, SELECT retrieves records, UPDATE modifies existing records, and DELETE removes records. These operations are used to manage the database.

### Database design

The database is divided into Student, Subject, Examination, Marks and Result tables to keep the data organized and reduce unnecessary duplication. Foreign keys connect the related tables.

### Keys and Constraints

Primary keys uniquely identify records. Foreign keys establish relationships between tables. NOT NULL ensures required values are entered, while UNIQUE prevents duplicate values

### Aggregate Functions and Joins

COUNT, SUM, AVG, MAX, and MIN perform calculations on groups of records. INNER JOIN combines related information from Customer, Vehicle, and Service tables using their keys.

## 6.Conclusion - 5 Marks

### Future Enhancements

- Student login and authentication
- Onlin eexaminationschedulingandnotifications.
- Automated resultpublication.
- Printablem arksheetsandresultreports.
- Attendance integration.
- Faculty/admindashboardforexaminationmanagement.
- Integration withawebormobileapplication.
- Automated databasebackupandrecovery.

### Conclusion

T he student Examination database provides a stuctural way to manage student details,subject,examination records marks and results.Primarykeys, foreignkeys, constraints, SQL operations,filtering,grouping,gradecalculation and INNERJOIN were used to implement the required database functions.Theproject demonstrates practica lunderstanding of relational database design and SQL-based examination data management.